C455/B503 Algorithm Analysis
Spring 2026 Syllabus

Description: Models, algorithms, recurrences, summations, growth rates. Probabilistic tools, upper and lower bounds; worst-case and average-case analysis, amortized analysis, dynamization. Comparison-based algorithms: search, selection, sorting, hashing. Information extraction algorithms (graphs, databases). Graphs algorithms: spanning trees, shortest paths, connectivity, depth-first search, breadth-first search.

Prerequisite: C343/A594, C250/C251/C241, and M260 Probability or Statistics or M216/M230 Calculus 2 or equivalent.

Instructor:	Dr. Dana Vrajitoru	Office:	NS 337
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url:	http://www.cs.iusb.edu/~dvrajito/		
Course web page:	http://www.cs.iusb.edu/~dvrajito/teach/c455/		
Office hours:	MW 2 – 4 pm, TR 3 – 5 pm, or by appointment.		
Zoom link:	https://iu.zoom.us/j/81040608225		

Textbook: *Practical Analysis of Algorithms* by D. Vrajitoru, W. Knight, Springer, 2014.

Required: Computer with any operating system, Google Chrome browser, camera, microphone.

Grading system:

Weekly homework assignments	20 points each
Weekly quizzes	5 points each
Midterm exam	50 points
Final exam	50 points

Final grade:

Letter	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
Percentage	97	93	90	87	83	80	77	73	70	67	63	60	0

Guidelines for assignments:

- The assignments will be posted on Canvas and must be submitted on Canvas.
- The assignments are due at midnight of the due date.
- Assignments written on paper can be scanned and submitted as pdfs or MS Word documents. Each homework submission must be submitted as a single pdf or docx file, not as multiple image files.
- Resubmissions are accepted with applicable late penalty. The latest submission will be the one graded.
- No homework accepted after 1 week from the due date. A late homework loses 10% of the points per day, up to 50% of the points.

- For problems that have a short answer, such as a number, typically, a detailed explanation of the steps taken to obtain that answer is expected.
- Weekly quizzes are neither proctored nor timed. The deadline for them is strict; no grace period after the deadline.
- The midterm and final exams will be proctored by Honorlock 1.3 and timed at 2 hours. You will need to use the Chrome browser and install an extension for it (details will be available). You will need to turn on your camera and microphone during the exam, so you will need access to this equipment.
- All of the assignments are individual. Consulting with colleagues is acceptable, but homework submissions that are too similar can be penalized or rejected. No credit will be given for homework assignments obtained from external sources unless explicitly allowed.
- See separate guideline for the achievements.

Course Objectives:

Upon completion of this course, students will be able to do the following:

- Analyze algorithms and estimate their best-case, worst-case, and average-case behavior.
- Compare different algorithms solving the same problem and choose the most efficient one.
- Demonstrate a familiarity with major algorithms and data structures.
- Apply important algorithmic design paradigms and methods of analysis.

More Information:

Please consult the following link for the most up to date information on COVID-19 measures, accessibility and accommodations, academic/technology support centers, academic honesty, sexual misconduct, and Title IX:
https://academics.iusb.edu/syllabus-statements/index.html

Campus Spring 26 calendar:
https://southbend.iu.edu/students/registrar/calendars/2026-spring.html

Course Outline

Week	Chapter	Homework, Exams	Due Date
1	Math review, Big O, Ω, and Θ Notations	Homework 1, Quiz 1	Jan 21
2	MLK Jr. Holiday		Jan 19
	Little oh Notation, Big O, Ω, and Θ Notations	Homework 2, Quiz 2	Jan 28
3	Asymptotic Functions, Floor and Ceiling, Logarithms,	Homework 3, Quiz 3	Feb 4
4	Sums and Sum Notation, Factorial, Remainders in Integer Division	Homework 4, Quiz 4	Feb 11
5	Recurrence Relations, Definition Linear RR with constant coefficients, Linear RR nonhomogeneous polynomial	Homework 5, Quiz 5	Feb 18
6	Linear RR nonhomogeneous exponential RR depending on floor(n/2) and ceil(n/2)	Homework 6, Quiz 6	Feb 25
7	Deterministic AA, Loop counting, Euclid's algorithm for the GCD	Midterm	Mar 4
8	Divide and Conquer Algorithms, Binary Trees, Binary Search Trees	Homework 7, Quiz 7	Mar 11
9	Sorting theorem, Heap, Priority Queue	Homework 8, Quiz 8	Mar 25
	SPRING BREAK		Mar 15-21
10	Pattern Matching Algorithms, Randomizing data, Probability Review	Homework 9, Quiz 9	Apr 1
11	Randomized algorithms, simulating observations	Homework 10, Quiz 10	Apr 8
12	Expected/average behavior of algorithms	Homework 11, Quiz 11	Apr 15
13	Graph theory	Homework 12, Quiz 12	Apr 22
14	Graph theory	Homework 13, Quiz 13	Apr 29
15	Review		
Final		Final	May 6

Academic Honesty Statement

Academic honesty is fundamental to the activities and principles of this university. It is the responsibility of the student to know prohibited actions such as cheating, fabrication, plagiarism, interference, violation of coursework, facilitating misconduct, and personal misconduct, and take proactive steps to avoid them. All students are held to the standards outlined in the Indiana University Student Code of Conduct found at https://policies.iu.edu/policies/stu-00-student-code/index.html. The academic community regards academic dishonesty as an extremely serious matter, with consequences that range from but are not limited to: receiving a warning, reduction in points, failing the assignment or the course, disciplinary probation, suspension and/or expulsion from the University.

Accessibility and Accommodations Statement

Indiana University is dedicated to ensuring that students with disabilities have the support services and reasonable accommodations needed to provide equal access to academic programs. To request an accommodation, you must establish your eligibility by working with Accessible Educational Services (AES) on your campus (sbdss@iu.edu).

Additional information can be found at accessibility.iu.edu. Note that services are confidential, may take time to put into place, and are not retroactive; captions and alternate media for print materials may take three or more weeks to get produced. Please contact your campus AES office as soon as possible if accommodations are needed.

Additionally, IU is dedicated to supporting students who are pregnant or have experienced other conditions related to pregnancy (termination of pregnancy, miscarriage, lactation, or related medical conditions). To register for accommodations due to pregnancy or a related medical condition, please contact our campus Accessible Educational Services (AES) office at sbdss@iu.edu. To learn about the rights and resources available to students, such as academic accommodations, please visit pregnancy.iu.edu or email ocrc@iu.edu for more information.

Accommodations for Religious Observances Statement

Indiana University respects the right of all students to observe their religion. If you will require academic accommodations for a religious observance, please provide the instructor a request by the end of the second week of the course. If the instructor and student agree on the accommodation, the plan should be carried out. If after discussion, there is no consensus on the accommodation, either party or both may ask advice of the Department Chair or the Dean, and if no consensus is reached, then the advice of the Executive Vice Chancellor of Academic Affairs (EVCAA). The student or instructor may bring any questions or concerns to the Office of Civil Rights compliance by emailing ocrc@iu.edu or may appeal any campus decisions with that office.

Indiana University Policy on Discrimination, Harassment, and Sexual Misconduct

As your instructor, one of my responsibilities is to create a positive learning environment for all students. IU policy prohibits sexual misconduct in any form, including sexual harassment, sexual assault, stalking, sexual exploitation, and dating and domestic violence. If you have experienced sexual misconduct, or know someone who has, the University can help. If you are seeking help and would like to speak to someone confidentially, you can contact [insert specific campus office and contact info]. Additionally, Indiana University also prohibits discrimination, see the university's Non-Discrimination policy here: https://policies.iu.edu/policies/ua-01-non-discrimination/index.html

It is also important that you know that because of my role, University policy requires me to share information brought to my attention about potential sexual misconduct. In that event, you may contacted with information about resources and your options for any next steps. Protecting student privacy is of utmost concern, and information will only be shared with those that need to know to ensure the university can respond and assist. I encourage you to visit stopsexualviolence.iu.edu and ocrc.iu.edu to learn more.

If you have experienced sexual misconduct, or know someone who has, the University can help. If you are seeking help or would like to report sexual misconduct, you can contact Josie Brush, Deputy Sexual Misconduct & Title IX Coordinator for Regional Campuses, 812-855-2751, jvbrush@iu.edu.